Response-aware path compression

Sharp planar distortion and exact Euler storage

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Abstract

Path compression aims to store a driving signal economically while preserving its effect on differential equations. This discussion draft considers linear systems with a fixed joint Frobenius-norm bound on their coefficient matrices. We measure approximation error by the worst change in terminal response, and compare it with the weighted Hilbert–Schmidt error between the path's signature and its decoded approximation. This tensor error provides a sufficient certificate of response accuracy, but may be conservative.

For planar bounded-variation paths decoded by the degree-two log-ODE method, we prove that the response-to-certificate ratio tends to $2\sqrt{2}/9$ as the path is scaled down, on a fixed partition with nonzero cubic error. The limit is uniform in response dimension $n \geq 2$. We obtain explicit finite-scale and perturbation bounds, and prove separation at every positive scale for chains of adjacent area-only blocks. Separately, for Euler decoding, we identify the exact worst-case bounded-variation error when the partition may depend on the path but not on the system. Using this constraint instead of a length-based upper bound can change the best choice of truncation depth and interval count, reducing finite storage costs without changing their leading asymptotics. These results clarify when certifying a representation's tensor error is more demanding than preserving its driven responses.

1 What is compressed, and what must be preserved?

Suppose a path is used to drive a differential equation. We want to replace the recorded trajectory by a smaller collection of numbers, while retaining enough information to reproduce the equation's response. The goal is not necessarily to reconstruct a visually similar path: it is to control what the driven system can detect. We consider terminal responses of linear systems with bounded coefficients, following the criterion of Ferrucci, Perrée and Lyons (FPL) [1].

1.1 Stored data and the decoder

Let $X : [0,1] \to \mathbb{R}^d$ be a continuous bounded-variation path. Split it at $\pi = (0 = t_0 < \cdots < t_m = 1)$, and call its restriction to interval i X_i. The signature $S(X_i)$ collects its iterated integrals; degree j consists of integrals with j successive increments. A compressed representation stores *one truncated log-signature per interval*. The truncation depth controls how much is kept on each interval; the interval count controls how many such records are stored.

Our first comparison keeps degree two. In the plane, this retains displacement and signed area, but discards higher-order information about the local path. Write

$$\ell_i^{\leq 2} = \ell_{i,1} + \ell_{i,2}, \qquad \ell_{i,j} = (\log_\otimes S(X_i))_j.$$

Here i indexes intervals and j indexes tensor degree. Exponentiating each stored log and composing in time order produces the decoded tensor series

$$F = S(X), \qquad G_\pi = \prod_{i=1}^{m} \exp_\otimes(\ell_i^{\leq 2}).$$

These are the two objects compared below: F is exact, whereas G_π is reconstructed from the finite stored data. Although G_π is an infinite series, its coefficients need not be stored individually.

To turn either series into a response, fix real $n \times n$ matrices $A = (A_1, \dots, A_d)$. There are d driver channels and n response states. A row-state linear system satisfies

$$dY = \sum_{i=1}^{d} Y A_i \, dX^i, \qquad Y_0 = I_n.$$

Its terminal response is obtained by replacing each tensor letter by its matrix. If e_i is the i-th driver basis vector, define $R_A(e_{i_1} \otimes \cdots \otimes e_{i_j}) = A_{i_1} \cdots A_{i_j}$, extended linearly and degree by degree. Brackets denote commutators in the tensor or matrix algebra; the empty word evaluates to I_n. The exact and decoded responses are therefore

$$\Phi^A(X) = R_A(F), \qquad \widehat{\Phi}^A_\pi(X) = R_A(G_\pi) = \prod_{i=1}^{m} \exp\bigl(R_A(\ell_i^{\le 2})\bigr).$$

Thus decoding means evaluating a stored Lie polynomial, taking a matrix exponential, and composing the interval responses. The same stored data can be used for different systems; the same tuple A acts on every interval. Transposing tuples gives the column-state convention with unchanged norms.

$$X \xrightarrow{\text{solve exactly}} \Phi^A(X),$$

$$X \xrightarrow{\text{split } \pi} \underbrace{(\ell_i^{\le 2})_{i=1}^{m}}_{\text{finite stored data}} \xrightarrow{\text{exponentiate and compose}} \widehat{\Phi}^A_\pi(X).$$

1.2 True response error versus a tensor certificate

To ask for accuracy across a family of systems, impose a shared coefficient budget $|A|^2 = \sum_i \|A_i\|_F^2 \le r^2$, where $\|\cdot\|_F$ is the Frobenius norm and $r > 0$. The *true response error* is the worst terminal discrepancy over that family:

$$\Delta_{r,n}(F, G) = \sup_{|A| \le r} \|R_A(F) - R_A(G)\|_{\mathrm{op}}. \tag{1}$$

The Euclidean operator norm controls the error for every unit initial state. For the degree-two representation, use $G = G_\pi$.

A simpler sufficient bound compares tensor coefficients before choosing a system. Let F_j, G_j denote degree-j components, with standard word tensors orthonormal. Their Hilbert–Schmidt (HS) norm is the Euclidean norm of the list of word coefficients. The weighted certificate is

$$\mathcal{B}_r(F, G) = \sum_{j \ge 1} r^j \, \|F_j - G_j\|_{\mathrm{HS}}.$$

The factor r^j accounts for products of j bounded matrices. Cauchy–Schwarz gives $\Delta_{r,n} \le \mathcal{B}_r$; both series converge in the classes used here. For a tolerance ε, $\Delta_{r,n} \le \varepsilon$ means that every admissible system is accurate, while $\mathcal{B}_r \le \varepsilon$ is a sufficient way to certify that accuracy. Our question is whether this sufficient test rules out compressed representations whose responses are already accurate enough.

There is no universal discount on the certificate. For example, comparing a positive straight-line signal with the zero signal, a scalar or rank-one matrix response attains the HS bound. The separation below instead exploits the bracket structure of the discarded terms. Sections 2–3 use log-ODE decoding; Section 4 considers the distinct Euler decoder.

2 Planar log-ODE error: the first discarded term

Now take $d = 2$, keeping the partition π fixed. On each interval, the exact signature and the decoded series agree through degree two. The first possible difference is cubic, and these first differences add across intervals:

$$D_3 := (F - G_\pi)_3 = \sum_i \ell_{i,3}.$$

This is the part of the local log-signatures first omitted by the stored data. If it is nonzero, it sets the leading size of both errors. The issue is how much of that tensor error a bounded matrix system can actually realise.

For a small-scale comparison, replace X by sX while keeping the coefficient budget r fixed. Scaling the path multiplies tensor degree j by s^j; denote this operation by δ_s. Path scale and coefficient budget combine into $t = sr$. It is convenient to use a unit budget and write

$$\Delta_n(t) = \Delta_{1,n}(\delta_t F, \delta_t G_\pi), \qquad \mathcal{B}(t) = \mathcal{B}_1(\delta_t F, \delta_t G_\pi).$$

These equal the response error and certificate for the representation of sX at budget r. In the nonzero-cubic case, the next theorem compares their leading terms of order t^3.

Theorem 1. *If $D_3 \neq 0$, then for every $n \geq 2$,*

$$\frac{\Delta_n(t)}{\mathcal{B}(t)} \longrightarrow \rho := \frac{2\sqrt{2}}{9} \quad (t \downarrow 0),$$

uniformly in n. There is also an explicit positive scale radius on which the response error is strictly smaller than its HS certificate. This separation persists on a bounded-variation perturbation neighborhood of X.

Why this constant? The HS certificate measures word coefficients in an orthonormal tensor basis. A response cannot choose those coefficients' matrix products independently: all products come from one shared, budget-constrained tuple. For a cubic Lie error they combine into nested commutators. After a driver rotation, every nonzero planar cubic direction is a multiple of $[e_1, [e_1, e_2]]$. Its three word coefficients are $1, -2, 1$, so its HS norm is $\sqrt{6}$. Two Böttcher–Wenzel inequalities [2], together with the joint unit budget, give a sharp matrix-response maximum $4/(3\sqrt{3})$, attained already in dimension two. Their quotient is

$$\frac{4/(3\sqrt{3})}{\sqrt{6}} = \frac{2\sqrt{2}}{9}.$$

Thus the certificate overestimates the leading response error by about 3.18, not because all tensor errors admit the same improvement, but because this leading error has a specific commutator structure. Appendix A gives the attainment argument and explicit tail and perturbation estimates; their auxiliary constants are not needed to read the comparison.

A finite compression decision. Take the corner $0 \to e_1 \to e_1 + e_2$, and store displacement and area as one interval. With unit coefficient budget, the exact and decoded responses of the scaled path are

$$\Phi^A(tX) = e^{tA_1} e^{tA_2}, \qquad \widehat{\Phi}^A_\pi(tX) = e^{t(A_1+A_2)+(t^2/2)[A_1,A_2]}.$$

At $t = 1/20$, rational enclosures prove, for every $n \geq 2$,

$$\Delta_n(t) < 1.934 \times 10^{-5} < 2 \times 10^{-5} < 3.608 \times 10^{-5} < \mathcal{B}(t). \tag{2}$$

Tolerance 2×10^{-5} therefore accepts the one-interval representation under a proved response bound, but rejects it under the full HS certificate. This is a finite-scale decision, not only a limiting ratio or a numerical optimizer's output. Appendix F supplies the exact arithmetic.

The fixed partition and $D_3 \neq 0$ are essential. A two-interval square loop can have zero cubic discrepancy but nonzero quartic error. Refining the partition can also make the cubic term vanish. Theorem 1 supplies no mesh-uniform conclusion or extension to general rough input with $2 < p < 3$.

3 Homogeneous merges at every positive scale

The previous result compares a path with its local degree-two approximation. We now ask whether several stored log-signatures can themselves be replaced by one. Consider area-only blocks in consecutive coordinate planes. The fine representation keeps their order; the merged representation stores only their sum. Under a matrix response, these area blocks need not commute, so merging can change the response even though the retained total area is unchanged.

For a chain of $\ell \geq 3$ blocks, in driver dimension $d = \ell + 1$, let $U_i = [e_i, e_{i+1}]$ and compare decoded depth-two series

$$F_\lambda = \prod_{i=1}^{\ell} e_\otimes^{\lambda U_i}, \qquad G_\lambda = e_\otimes^{\lambda \sum_i U_i}, \qquad \lambda > 0.$$

Here U_i records area in the $(i, i+1)$ coordinate plane, and λ is its amplitude. This is distinct from the path scale t in Section 2. Use unit tuple budget in $\Delta_n(\lambda) = \Delta_{1,n}(F_\lambda, G_\lambda)$ and $\mathcal{B}(\lambda) = \mathcal{B}_1(F_\lambda, G_\lambda)$.

Theorem 2. *For every $\ell \geq 3$, every response dimension $n \geq 1$ and $\lambda > 0$,*

$$\Delta_n(\lambda) < \frac{\lambda^2}{\sqrt{2}} e^{\sqrt{2}\lambda} < \frac{e^{\sqrt{6}\lambda} - 1 - \sqrt{6}\lambda}{\sqrt{3}} \leq \mathcal{B}(\lambda). \tag{3}$$

The separation bounds are independent of chain length.

This permits an ℓ-to-one merge at a nonempty interval of response tolerances rejected by the full certificate. With generic depth-two log-signature layout held fixed, storage falls by factor ℓ. The theorem controls the additional error caused by merging these decoded blocks. If they came from truncating actual BV loops, the fine representation is already an approximation: its error must also be included in the total error relative to the original path. For pure area rough blocks, the fine series is exact and there is no initial decoding error to add. Appendix B verifies the infinitely many coefficient comparisons analytically; checking finitely many coefficients alone would not prove the all-scale assertion.

4 Exact Euler minimax and storage

The third comparison is a storage optimization problem, not a consequence of multiplying a general error bound by the planar factor $2\sqrt{2}/9$. Choose the truncation depth N and interval count m to minimize storage while meeting a response tolerance. Here the decoder is *Euler*: instead of exponentiating each truncated log-signature, it evaluates a truncated signature polynomial. Log-signature coordinates may still be used for storage; the required signature levels are reconstructed from them before decoding.

For degree-N Euler decoding on m intervals, put $P_N(x) = \sum_{j=0}^{N} x^j/j!$. Write $R_{A,j}$ for the restriction of R_A to degree j. The decoded response is

$$E^A_{N,\pi}(X) = \prod_i \left[I + \sum_{j=1}^{N} R_{A,j}(S(X_i)_j) \right].$$

We ask for the worst error over all paths of variation at most one and all coefficient tuples with budget r. For each path we may choose its partition, but that partition must work for every admissible tuple. The order of optimization in the next theorem makes this requirement explicit.

Theorem 3. *For any $N, m \geq 1$, fixed driver and response dimensions, coefficient radius $r > 0$, and paths of variation at most one,*

$$\sup_X \inf_{\pi:\, m \text{ intervals}} \sup_{|A| \leq r} \left\| \Phi^A(X) - E^A_{N,\pi}(X) \right\|_{\mathrm{op}} = \mathcal{E}(N, m, r) := e^r - P_N(r/m)^m. \tag{4}$$

Partitions may depend on X, but not on A. Equal-variation partitions give the upper bound; a straight line with a scalar/rank-one development attains the minimax error.

FPL already display the scalar expression in their equation (1.13). Appendix C proves its universal majorant and partition minimization. This refines the additional length envelope

$$B(N, m, r) = \frac{e^r r^{N+1}}{(N+1)! m^N}.$$

For a fair storage comparison, hold the decoder and coordinate layout fixed. Let $\Lambda_d(N)$ be the number of independent log-signature coordinates through degree N in d channels. The dense generic representation stores $m\Lambda_d(N)$ scalars. Appendix D gives the dimension formula and the limiting optimization arguments.

Finite integer movement. For $d = 2, r = 1, \varepsilon = 10^{-2}$, the two optima are

Constraint	N	m	stored scalars
Exact BV minimax	4	1	8
Length envelope	5	1	14

Appendix F certifies global minimality with rational arithmetic. On the retained grid $d \in \{2, 3, 5\}$, $r \in \{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$ and $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-4}, 10^{-8}, 10^{-12}\}$, optimum sets change in 85 of 135 cases; 84 have strictly lower storage, with largest saving 42.9%. These are finite-grid observations. Integer searches use high-precision boundary refinement; the general grid is not interval certified.

What changes, and what does not. The exact constraint can select a different pair of depth and interval count: in the displayed case, depth falls from five to four while the interval count stays at one. This is a finite-cost refinement, not a change to the leading BV storage rates. At fixed driver dimension $d \geq 2$, in either limit $\varepsilon \downarrow 0$ at fixed $r > 0$, or $r \to \infty$ at fixed $\varepsilon > 0$, the ratio of the two minimum costs tends to one. Appendix D states the depth, interval and cost asymptotics precisely.

A separate structured example uses pure higher-degree Engel directions, formed by repeatedly bracketing one driver with another. Their response can be much smaller than their HS certificate, and their HS-certified storage overhead can grow with amplitude under Euler decoding. This remains a decoder-specific comparison: a single pure log-signature block is exact under log-ODE decoding. Appendix E gives the response-storage bracket and its hypotheses.

5 Prior work and boundaries

Development separation through polynomial identities is already explicit in Chevyrev–Lyons, §4.2 and Theorem 4.8 [3]. The planar degree-six matrix identity is classical [4]. It illustrates a limitation of fixed-dimensional responses: attaining several low-degree maxima in dimension two does not mean that dimension two suffices for every Lie direction. Spectral Lie-polynomial evaluations and low-degree calculations have direct precedent in Boedihardjo–Geng–Souris [5], with different norm budgets; exact pure-path tail asymptotics appear in Li–Gao [6]. The Engel gauge below is a Böttcher–Wenzel corollary [2]. The product-formula representation is standard [7], as is fixed-system adaptive log-ODE analysis [8]. FPL's discussion after (2.10) already identifies the first discarded local log-signature error.

The candidate addition is the constrained response/HS comparison and its finite-scale consequences. General rough inputs with $2 < p < 3$, mesh-uniform nondegeneracy, exact finite-scale response suprema and general cubic-zero behavior remain open. The displayed results give no correction to FPL's general rough-path asymptotics or its empirical averaged-error figures.

A Planar gauge, tails and stability

Signature degree j is $S(X)_j = \int_{0<u_1<\cdots<u_j<1} dX_{u_1} \otimes \cdots \otimes dX_{u_j}$. For homogeneous D of degree j, write $\gamma_{j,n}(D) = \sup_{|A|\le 1} \|R_A(D)\|_{\mathrm{op}}$. Let $Z = [e_1, e_2]$. Every planar cubic Lie element is $D = [w, Z]$ with $w \in \mathbb{R}^2$, and $\|D\|_{\mathrm{HS}} = \sqrt{6}\,\|w\|_2$. Proper driver rotations preserve the tuple ball and fix Z, so reduce to $D = \|w\|_2\,[e_1, Z]$. Böttcher–Wenzel and scalar maximization give

$$\|[A_1, [A_1, A_2]]\|_{\mathrm{op}} \le 2\,\|A_1\|_F^2\,\|A_2\|_F \le \frac{4}{3\sqrt{3}}.$$

Equality holds at

$$A_1 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad A_2 = \frac{1}{\sqrt{3}}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

whose cubic response is $4E_{12}/(3\sqrt{3})$. Embedding proves equality for every $n \ge 2$; hence $\gamma_{3,n}(D) = \rho\,\|D\|_{\mathrm{HS}}$. One commutator step similarly gives $\gamma_{2,n}(\alpha Z) = |\alpha|/\sqrt{2} = \|\alpha Z\|_{\mathrm{HS}}/2$. It is attained at $A_1 = \mathrm{diag}(1,-1)/2$ and $A_2 = E_{12}/\sqrt{2}$.

Explicit tails and radius. Let L be the Euclidean variation of the planar path, and put

$$h = \|D_3\|_{\mathrm{HS}}\,, \qquad a = \sum_i \|\ell_{i,1}\|_2\,, \qquad b = \sum_i \|\ell_{i,2}\|_{\mathrm{HS}}\,.$$

The degree-four-and-above tail of the scalar majorant is

$$T_4(a,b;t) = e^{at+bt^2} - 1 - at - (a^2/2 + b)t^2 - (a^3/6 + ab)t^3.$$

All of its coefficients are nonnegative. Define $Q_R(t) = T_4(L, 0; t) + T_4(a, b/2; t)$ and $Q_H(t) = T_4(L, 0; t) + T_4(a, b; t)$. Then

$$|\Delta_n(t) - \rho h t^3| \le Q_R(t), \qquad ht^3 \le \mathcal{B}(t) \le ht^3 + Q_H(t). \tag{5}$$

When $h > 0$, choose any reference $T > 0$, set $C_T = Q_R(T)/T^4$, and let

$$t_* = \min\{T, (1-\rho)h/(2C_T)\}, \qquad c = (1+\rho)/2. \tag{6}$$

For $0 < t \le t_*$, these bounds give $\Delta_n(t) \le cht^3 < ht^3 \le \mathcal{B}(t)$. The following arguments prove the tails and their perturbation stability.

First discrepancy. Group-like series F, G agreeing through degree two have $D_3 = F_3 - G_3 = (\log F)_3 - (\log G)_3$, a cubic Lie tensor. The cubic formula and uniform response tail yield

$$\rho h t^3 - Q_R(t) \le \Delta_n(t) \le \rho h t^3 + Q_R(t).$$

The certificate includes its degree-three term exactly, giving (5). Thus

$$\left|\frac{\Delta_n(t)}{\mathcal{B}(t)} - \rho\right| \le \frac{Q_R(t) + \rho Q_H(t)}{ht^3}.$$

Both tails are $O(t^4)$, independently of n. For partitioned degree-two log-ODE, interval factors agree through degree two and first differ by $\ell_{i,3}$. Multiplication by any nonconstant factor raises degree, so $D_3 = \sum_i \ell_{i,3}$.

Full series control. BV factorial bounds majorize F by e^{Lt}. Tensor products majorize G_π by e^{at+bt^2}. The sharp area gauge majorizes the matrix decoder by $e^{at+(b/2)t^2}$. Removing degrees zero to three yields Q_R, Q_H. For numerical evaluation, the positive identity

$$T_4(a,b;t) = e^p - P_3(p) + q(e^p - P_1(p)) + e^p(e^q - P_1(q)), \quad p = at,\ q = bt^2,$$

avoids small-scale subtractive cancellation. Nonnegative coefficients imply $Q_R(t) \le C_T t^4$ on $0 \le t \le T$, which proves (6) and the strict gap.

BV stability. For anchored paths of length at most L and $\eta = \|X - Y\|_{1\text{-var}}$, interpolation and insertion of one $d(Y - X)$ into the iterated integral give

$$\|S_j(X) - S_j(Y)\|_{\mathrm{HS}} \le \frac{\eta L^{j-1}}{(j-1)!}.$$

At the insertion time, summing over positions produces $(L_{\text{before}} + L_{\text{after}})^{j-1}/(j-1)!$; integrate against the variation measure of $Y - X$, then along interpolation. In

$$\ell_3 = S_3 - \tfrac{1}{2}(S_1 \otimes S_2 + S_2 \otimes S_1) + \tfrac{1}{3}S_1^{\otimes 3},$$

the three differences have bounds $\eta L^2/2$, $3\eta L^2/2$, and ηL^2. Therefore $\|\ell_3(X) - \ell_3(Y)\|_{\mathrm{HS}} \le 3\eta L^2$. On a fixed common partition, local difference variations sum to η; the same bound applies to D_3. Also $a \le L$, $b \le \sum_i L_i^2 \le L^2$. For a baseline of length L_X and cubic norm $h_X > 0$, set

$$L = L_X + 1, \qquad \eta_* = \min\{1, h_X/(6L^2)\}.$$

Perturbations within η_* have cubic norm at least $h_X/2$. Use $Q_R^*(t) = T_4(L, 0; t) + T_4(L, L^2/2; t)$ and this lower norm in (6) for a common positive radius.

Mixed blocks. For increments v_i and areas $\alpha_i Z$, compare

$$e_{\otimes}^{tv_1+t^2\alpha_1 Z} e_{\otimes}^{tv_2+t^2\alpha_2 Z} \quad \text{with} \quad e_{\otimes}^{t(v_1+v_2)+t^2(\alpha_1+\alpha_2+\omega/2)Z}, \qquad \omega = \det(v_1, v_2).$$

BCH at degree three gives $[w, Z]$, where

$$w = \tfrac{1}{2}(\alpha_2 v_1 - \alpha_1 v_2) + \tfrac{\omega}{12}(v_1 - v_2).$$

The same lemma applies when $w \neq 0$, with response tails $T_4(a_f, b_f/2; t) + T_4(a_c, b_c/2; t)$, where

$$a_f = \|v_1\| + \|v_2\|, \quad b_f = \sqrt{2}(|\alpha_1| + |\alpha_2|), \quad a_c = \|v_1 + v_2\|, \quad b_c = \sqrt{2}|\alpha_1 + \alpha_2 + \omega/2|.$$

For independent v_1, v_2, $w = 0$ exactly when $\alpha_1 = \alpha_2 = -\omega/6$. Pure planar area blocks commute.

B Gram certificates and all-scale chain proof

Let same-layer blocks $U_1, \ldots, U_\ell$ have Gram matrix $g_{ij} = \langle U_i, U_j \rangle_{\mathrm{HS}}$. For a weak composition α of m, let $w(\alpha)$ be its nondecreasing word, $\alpha! = \prod_i \alpha_i!$, and $W_\alpha = U_1^{\otimes \alpha_1} \otimes \cdots \otimes U_\ell^{\otimes \alpha_\ell}$. The meta-degree terms are

$$F_m = \sum_{|\alpha|=m} W_\alpha/\alpha!, \qquad G_m = (\sum_i U_i)^{\otimes m}/m!.$$

With $H_m = m!\,\|F_m - G_m\|_{\mathrm{HS}}$, tensor inner products give

$$H_m^2 = \sum_{|\alpha|=|\beta|=m} \binom{m}{\alpha}\binom{m}{\beta} \prod_{v=1}^{m} g_{w(\alpha)_v, w(\beta)_v} - \left(\sum_{i,j} g_{ij}\right)^m. \tag{7}$$

Indeed $\langle W_\alpha, W_\beta \rangle$ is the displayed product. Writing $s_i = \sum_j g_{ij}$, the multinomial theorem gives $\langle F_m, G_m \rangle = (\sum_i s_i)^m/(m!)^2 = \|G_m\|_{\mathrm{HS}}^2$, which proves (7).

Response bound. For $X_i = R_A(U_i)$ on the unit tuple ball, put

$$M = \sup_A \sum_i \|X_i\|_{\mathrm{op}}, \qquad K = \sup_A \sum_{i<j} \|[X_i, X_j]\|_{\mathrm{op}}.$$

If $M = 0$, all block responses vanish. For chains the positive certificate bound below proves separation directly; otherwise assume $M > 0$. The usual two-factor commutator representation [7] gives

$$\left\| e^{\lambda X} e^{\lambda Y} - e^{\lambda(X+Y)} \right\|_{\mathrm{op}} \le \frac{\lambda^2}{2} \|[X,Y]\|_{\mathrm{op}} e^{\lambda(\|X\|_{\mathrm{op}} + \|Y\|_{\mathrm{op}})}.$$

For completeness, differentiate $e^{(\lambda-s)(X+Y)} e^{sX} e^{sY}$ and use

$$[e^{sX}, Y] = \int_0^s e^{(s-u)X} [X,Y] e^{uX}\, du.$$

Induction over factors, with $[\sum_{i<j} X_i, X_j] = \sum_{i<j} [X_i, X_j]$, yields

$$\Delta_n(\lambda) \le \frac{\lambda^2 K}{2} e^{\lambda M}.$$

Comparing power coefficients with $\mathcal{B}(\lambda) = \sum_{m \ge 2} \lambda^m H_m / m!$, a sufficient condition is

$$K < \inf_{m \ge 2} \frac{2 H_m}{m(m-1) M^{m-2}} \quad (M > 0).$$

This infimum is genuinely infinite.

Analytic verification for chains. For $U_i = [e_i, e_{i+1}]$, blocks are mutually orthogonal with norm $\sqrt{2}$. Let $a_i = \|A_i\|_F$ and $c_\ell = \cos(\pi/(\ell+2))$. The path-graph largest eigenvalue gives $\sum_i a_i a_{i+1} \le c_\ell \sum_i a_i^2 \le c_\ell$. Böttcher–Wenzel therefore gives

$$M \le \sqrt{2} c_\ell < \sqrt{2}, \qquad K \le 2\sqrt{2} \sum_{i<j} (a_i a_{i+1})(a_j a_{j+1}) \le \sqrt{2} c_\ell^2 < \sqrt{2}.$$

The K inequality is pointwise before taking its supremum. Retain only nonmonotone words over the first three orthogonal blocks. There are $3^m - \binom{m+2}{2}$ such words, each absent from F_m and having coefficient $1/m!$ in G_m. Hence

$$H_m^2 \ge 2^m \left(3^m - \binom{m+2}{2}\right) \ge 2^m 3^{m-1}, \qquad m \ge 2.$$

The counting inequality follows at $m = 2$ and by induction: $\binom{m+3}{2} / \binom{m+2}{2} = (m+3)/(m+1) < 3$. Consequently every coefficient threshold is at least

$$L_m = \frac{4\, 3^{(m-1)/2}}{m(m-1)}.$$

Its consecutive ratio is $\sqrt{3}(m-1)/(m+1)$: below one up to $m = 3$, and above one for $m \ge 4$. For $m = 4 + s$, the latter follows from $3(m-1)^2 - (m+1)^2 = 2s^2 + 8s + 2 > 0$. Thus $\inf_{m \ge 2} L_m = L_4 = \sqrt{3} > \sqrt{2} > K$. Summing $H_m \ge (\sqrt{6})^m / \sqrt{3}$ gives the last bound in (3); the same coefficient comparison gives its middle strict inequality. This proves Theorem 2 at every positive scale.

C Euler minimax proof

Put $x_i = |A|\,\|X_i\|_{1\text{-var}}$ and $R = \sum_i x_i$. Expanding all exact interval factors, the Euler product retains precisely multi-indices with every interval degree at most N. Factorial estimates give

$$\left\|\Phi^A(X) - E^A_{N,\pi}(X)\right\|_{\mathrm{op}} \le \sum_{\max j_i > N} \prod_i \frac{x_i^{j_i}}{j_i!} = e^R - \prod_i P_N(x_i).$$

The majorant increases in each x_i. For $N \ge 2$, direct coefficient comparison gives

$$P_{N-1}(x)^2 - P_N(x)P_{N-2}(x) = \frac{x^{N-1}}{N!}\sum_{j=0}^{N-1}(N-j)\frac{x^j}{j!} \ge 0.$$

For $N = 1$, $\log P_1 = \log(1+x)$ is concave as well. Thus $\log P_N$ is concave, and Jensen maximizes the product at equal $x_i = R/m$. An equal-variation partition is independent of A and gives the uniform bound $\mathcal{E}(N, m, r)$ on the normalized path class.

For $X_t = te_1$, take $A_1 = rQ$ with a rank-one orthogonal projection and other matrices zero. On its range the error is $e^r - \prod_i P_N(r\Delta t_i)$. Jensen shows every partition has error at least $\mathcal{E}(N, m, r)$; equal intervals attain it. This proves the quantifier order in (4).

D Storage limit arguments

The dense generic log-signature dimension is given by Witt's formula:

$$\Lambda_d(N) = \sum_{j=1}^{N} \frac{1}{j}\sum_{v|j} \mu(v) d^{j/v},$$

where μ is the Möbius function.

Limiting storage statement. Let $C_{d,k}(z, \varepsilon)$ minimize $m\Lambda_d(kq)$ under $\mathcal{E}(q, m, z) \le \varepsilon$, with fixed $d \ge 2, k \ge 1$. Here q, m are positive integers, and kq is the stored log-signature depth. Put $\beta = k \log d$, $\lambda = z + \log(z/\varepsilon)$. In either limit $z \to \infty$ at fixed $\varepsilon > 0$, or $\varepsilon \downarrow 0$ at fixed $z > 0$, any optimizer satisfies

$$q_* \sim \sqrt{\lambda/\beta}, \qquad \log(m_*/z) \sim \sqrt{\beta\lambda}, \qquad \log(C_{d,k}/z) \sim 2\sqrt{\beta\lambda}. \tag{8}$$

Replacing $\mathcal{E}$ by B changes minimum cost by a relative factor tending to one, as proved below. For $k = 1$, this preserves the leading BV storage asymptotics in FPL.

Proof. Write $x = z/m$, $a = e^{-x}\sum_{j>q} x^j/j!$, and $B = e^z z^{q+1}/((q+1)!m^q)$. Poisson-tail integral bounds, or repeated integration of e^{-u}, give

$$e^{-x}\frac{x^{q+1}}{(q+1)!} \le a \le \frac{x^{q+1}}{(q+1)!}.$$

Since $\mathcal{E} = e^z[1 - (1-a)^m]$, Bernoulli's inequality yields

$$\frac{e^{-x}}{1+ma} \le \frac{\mathcal{E}}{B} \le 1. \tag{9}$$

Feasibility $\mathcal{E} \le \varepsilon < e^z$ and $1 - (1-a)^m \ge 1 - e^{-ma}$ also imply

$$ma \le -\log(1 - \varepsilon e^{-z}). \tag{10}$$

Optimizer regime. For fixed d, k, Witt's formula gives $\log \Lambda_d(kq) = \beta q - \log q + O(1)$. Take q nearest $\sqrt{\lambda/\beta}$ and

$$m = \left\lceil z \exp\left(\frac{\lambda - \log((q+1)!)}{q}\right)\right\rceil.$$

This is B-feasible. Stirling and negligible relative rounding give $\log(C_{d,k}/z) \le 2\sqrt{\beta\lambda} + O(\log\lambda)$. Because $m \ge 1$, every optimizer has $q = O(\sqrt{\lambda})$. In the large-z regime, $\log z = O(\log\lambda)$ in this inference.

At fixed z, bounded m and $q = O(\sqrt{\lambda})$ would, by the first omitted term and $1-(1-a)^m \ge a$, give $\log\mathcal{E} \ge -O(\sqrt{\lambda}\log\lambda)$, contradicting $\log\varepsilon = -\lambda + O(1)$. Thus $m \to \infty$ and $x \to 0$. At $z \to \infty$, $m = 1$ gives $\log P_q(z) = O(\sqrt{z}\log z) = o(z)$, so $\mathcal{E} \sim e^z$, infeasible. If $m \ge 2$ and $x \ge c > 0$, then

$$\mathcal{E} \ge e^{z-x}\frac{x^{q+1}}{(q+1)!} \quad\Longrightarrow\quad \log\mathcal{E} \ge z/2 - O(\sqrt{z}\log z),$$

also infeasible. Hence $x \to 0$, and $m \to \infty$, in both regimes.

At fixed z, $ma \le zx^q/(q+1)! \le zx/2 \to 0$ once $x < 1$. For large z, (10) gives $ma \to 0$. It follows from (9) that $\mathcal{E}/B \to 1$ at every optimizer, so $\log B \le \log\varepsilon + o(1)$. Consequently

$$\log(C_{d,k}/z) \ge \beta q + \lambda/q - O(\log\lambda).$$

The minimum of $\beta q + \lambda/q$ is $2\sqrt{\beta\lambda}$. Its gap from that minimum forces $q/\sqrt{\lambda/\beta} \to 1$. Subtracting $\log\Lambda_d(kq)$ gives the formula for $\log(m/z)$.

Finally, at an exact optimizer inflate m by $\max\{1, (B/\varepsilon)^{1/q}\}$, rounding upward. This is B-feasible and costs a relative $1 + o(1)$ since $B/\varepsilon \le 1 + o(1)$ and $m \to \infty$. The exact feasible region contains the B-feasible one, proving convergence of the cost ratio to one.

E Engel gauge and structured storage scope

Pure Engel paths supply a separate restricted comparison. Fix driver dimension $d \ge 2$ and response dimension $n \ge 2$; the direction uses only the first two driver channels. For $k \ge 2$ and $E_k = \mathrm{ad}_{e_1}^{k-1}(e_2)$, set

$$h_k = \sqrt{\binom{2k-2}{k-1}}, \quad g_k = [2(k-1)]^{(k-1)/2}k^{-k/2}, \quad s_k = g_k/\sqrt{2}.$$

For a pure degree-k rough path, take $p \ge k$; effective amplitude $a = br^k > 0$ absorbs original amplitude b and coefficient radius r. Let $C_k^{\mathrm{true}}(a,\varepsilon)$ be the minimum dense generic cost $m\Lambda_d(N)$ over $N, m \ge 1$ meeting the true response tolerance, allowing path-adapted partitions independent of the coefficient tuple. When $\varepsilon < e^{as_k} - 1$, true Euler storage is bracketed by

$$C_{d,k}(as_k,\varepsilon) \le C_k^{\mathrm{true}}(a,\varepsilon) \le C_{d,k}(ag_k,\varepsilon) \le C_{d,k}(ah_k,\varepsilon).$$

Consequently, at fixed d, k, ε,

$$\liminf_{a\to\infty}\frac{1}{\sqrt{a}}\log\frac{C_{d,k}(ah_k,\varepsilon)}{C_k^{\mathrm{true}}(a,\varepsilon)} \ge 2\sqrt{k\log d}\,(\sqrt{h_k} - \sqrt{g_k}) > 0.$$

A single depth-k log-ODE factor represents this pure path exactly; the growing overhead concerns fixed Euler decoding and dense generic storage.

Gauges and storage proof. Repeated Böttcher–Wenzel gives

$$\left\|\mathrm{ad}_A^{k-1}(B)\right\|_{\mathrm{op}} \le 2^{(k-1)/2}\,\|A\|_F^{k-1}\,\|B\|_F\,.$$

Under $\|A\|_F^2 + \|B\|_F^2 \le 1$, the maximum occurs at squared norms $(k-1)/k$ and $1/k$, giving g_k. A diagonal block $A = \sqrt{(k-1)/(2k)}\,\mathrm{diag}(1,-1)$ and $B = E_{12}/\sqrt{k}$ attain operator norm g_k. Binomial expansion of E_k gives HS norm h_k. Its ratio g_k/h_k decays exponentially in k; this is a sharp corollary of a classical commutator inequality.

For the pure weakly geometric path $\exp_\otimes(taE_k)$, use $p \ge k$. Let $\delta_k(q,m,a)$ denote the worst unit-budget response error at Euler depth kq on the equal m-interval partition. Coefficientwise nonnegative error terms give

$$\mathcal{E}(q,m,as_k) \le \delta_k(q,m,a) \le \mathcal{E}(q,m,ag_k) < \mathcal{E}(q,m,ah_k).$$

For the lower witness, set $\alpha = \sqrt{(k-1)/k}$, $\eta = 1/\sqrt{k}$:

$$A_1 = \frac{\alpha}{\sqrt{2}}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \quad A_2 = \frac{\eta}{\sqrt{2}}\begin{pmatrix}0 & 1\\ (-1)^{k-1} & 0\end{pmatrix}.$$

Their unit tuple evaluates E_k to a symmetric matrix with off-diagonal entries s_k, hence positive eigenvalue s_k. Jensen gives the same witness lower bound for adaptive partitions. For the HS equality, each tensor-degree-kj error is $c_j a^j E_k^{\otimes j}$ with $c_j \ge 0$. Its norm is $c_j(ah_k)^j$, whose sum equals $\mathcal{E}(q,m,ah_k)$.

For $\varepsilon < e^{as_k} - 1$, depths below k are infeasible; intermediate depths have the same polynomial as the preceding multiple of k and larger storage. This gives the stated storage sandwich. The amplitude corollary follows from (8). At fixed a, all bracket endpoints instead have common leading logarithmic rate $2\sqrt{k\log d\log(1/\varepsilon)}$.

F Rational certificates and reproducibility

For the one-interval corner of Section 2, with $Z = [e_1, e_2]$,

$$D_3 = ([e_1,Z] - [e_2,Z])/12, \qquad h = \sqrt{3}/6, \quad a = \sqrt{2}, \quad b = \sqrt{2}/2.$$

For rational $x \ge 0$ and integer J with $x < J+2$,

$$P_J(x) \le e^x \le P_J(x) + \frac{x^{J+1}}{(J+1)!}\frac{1}{1 - x/(J+2)}.$$

This follows by a geometric upper bound on all successive omitted-term ratios. All displayed example enclosures use $J = 24$. Integer squares verify

$$\sqrt{2} < 707107/500000, \quad \sqrt{3} > 34641/20000, \quad \sqrt{6} < 244949/100000.$$

For the corner, $\rho h = \sqrt{6}/27$. Positive coefficients permit replacing $\sqrt{2}$ parameters by their upper bounds in Q_R; apply the exponential enclosure, and use the lower $\sqrt{3}$ bound in ht^3. Exact fractions then give (2).

For the storage example, bound $e = e^1$ by the same rational enclosure. The first five cumulative dimensions are $2, 3, 5, 8, 14$. Enumerating every (N, m) of cost at most 8 for $\mathcal{E}$, and at most 14 for B, gives the two unique optima in the table. Higher depths already cost more with $m = 1$. Each feasibility inequality is strictly decided by the rational bounds.

Source and computed examples are shipped in the accompanying source archive. Versioned source comprises the response modules, tests, runners and notebooks. The PDF is self-contained. The grid statistics are regenerated from retained CSV and checked against current integer optimization by the build runner. Ordinary floating-point plots/searches have no interval-certification claim.

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